# MATHEMATICS: THE NOT-SO-UNIVERSAL LANGUAGE

**Mathematical ideas cross languages. The words used to teach them can create extra challenges for multilingual learners.**

Five language features, illustrated with English examples and Spanish supports.

## 1 Comparatives

Words and structures that compare quantities or describe relationships.

| ENGLISH | WHAT IT MEANS | EN ESPAÑOL |
|---|---|---|
| **greater than** | Compare two quantities. | **mayor que** |
| **more than** | Start with y; add 5. | **más que** |
| **less than** | Start with y; subtract 5. | **menos que** |
| **twice / double** | Multiply by 2. | **el doble** |
| **at least / at most** | 10 or more / 10 or less. | **al menos / como máximo** |

## 2 Prepositions

Short connecting words that indicate relationships between quantities and operations.

| ENGLISH | WHAT IT MEANS | EN ESPAÑOL |
|---|---|---|
| **divided into** | How many groups of 4 fit into 9? | **dividido en** |
| **divided by** | 4 divided by 9 means 4 ÷ 9. | **dividido por** |
| **by** | Can indicate multiplication. | **por** |
| **exceeds ... by** | x is 7 greater than 2. | **excede a ... por** |

## 3 Passive Voice

Focuses on what happens to a quantity, without necessarily stating who acts.

| ENGLISH | WHAT IT MEANS | EN ESPAÑOL |
|---|---|---|
| **x is defined to be ≥ 0** | An inequality is given as a condition. | **se define como ...** |
| **15 is added to a number** | Find the number, then add 15. | **se le añade 15 ...** |
| **the expression is multiplied by 3** | Multiply the expression by 3. | **se multiplica por 3** |

## 4 Reversal Errors

Reversing the quantities when translating comparisons into equations.

| | | |
|---|---|---|
| **SUBTRACTION**<br>a is five less than b | **a = b − 5**<br>AVOID: a = 5 − b | **a es cinco menos que b** |
| **MULTIPLICATION**<br>five times as many students as professors | **s = 5p**<br>AVOID: 5s = p | **cinco veces tantos estudiantes como profesores** |

## 5 Logical Connectors

Words that express conditions and logical relationships.

| ENGLISH | WHAT IT MEANS | EN ESPAÑOL |
|---|---|---|
| **if ... then** | Condition and consequence. | **si ... entonces** |
| **if and only if** | Both directions must hold. | **si y solo si** |
| **given that** | Use a stated starting fact. | **dado que** |

**SOURCE:** Spanos, G., Rhodes, N. C., Dale, T. C., & Crandall, J. (2013). Linguistic features of mathematical problem solving: Insights and applications. In Linguistic and cultural influences on learning mathematics (pp. 221–240). Routledge.

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